

Foundations of Probability: Counting, Basic Probability, Conditional Probability & Bayes' Theorem

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Outline

- 1 Counting Principles
- 2 Probability Fundamentals
- 3 Conditional Probability
- 4 Total Probability and Bayes' Theorem

Counting Principles

The Mathematics of “How Many?”

1. The Multiplication Rule of Counting

Fundamental Counting Principle

If a task can be broken down into a sequence of k steps, where:

- Step 1 can be done in n_1 ways,
- Step 2 can be done in n_2 ways, ...,
- Step k can be done in n_k ways,

Then the total number of ways to perform the task is:

$$N = n_1 \times n_2 \times \cdots \times n_k$$

Multiplication Rule: Numerical Examples

Example 1: The ATM Pin

A bank card requires a 4-digit PIN (digits 0-9). How many unique PINs are possible?

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- Pos 1: 10 options (0-9)
- Pos 2: 10 options
- Pos 3: 10 options
- Pos 4: 10 options
- **Total:** $10 \times 10 \times 10 \times 10 = 10,000$ PINs.

2. The Addition Rule

Rule of Sum

If a task can be done in n_A ways OR in n_B ways, and these sets of ways are mutually exclusive (you cannot do both at once), then:

$$\text{Total ways} = n_A + n_B$$

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Example: Commute Options

You want to go to the office.

- You can take a **Bus** (3 different routes).
- OR you can take a **Train** (2 different lines).

You cannot take both simultaneously.

Total Choices: $3 + 2 = 5$ ways to get to work.

Factorials

Before we discuss Permutations, we must define the **Factorial**.

Definition

The factorial of a non-negative integer n , denoted by $n!$, is the product of all positive integers less than or equal to n .

$$n! = n \times (n - 1) \times (n - 2) \times \cdots \times 2 \times 1$$

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Recursion Rule: $n! = n \times (n - 1)!$

The Special Case of Zero: Why is $0! = 1$?

$$\text{Since } n! = n \times (n - 1)! \implies 1! = 1 \times 0! \implies 1 = 1 \times 0! \implies 0! = 1$$

Examples:

- $3! = 3 \times 2 \times 1 = 6$
- $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$

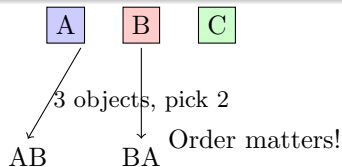
3. Permutations (Order Matters)

Key Intuition

Permutation is about **arrangement**. The order in which you place objects changes the outcome (e.g., Password “123” $\neq$ “321”).

Formula: The number of ways to arrange r objects chosen from n distinct objects:

$$P(n, r) = \frac{n!}{(n - r)!}$$



Permutations: Numerical Examples

Example: [Ranking] 10 participants in a race. How many ways can we award Gold, Silver, and Bronze?

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$$P(10, 3) = \frac{10!}{(7)!} = 10 \times 9 \times 8 = 720 \text{ ways.}$$

4. Combinations (Order Doesn't Matter)

Key Intuition

Combination is about **selection**. “Fruit Salad” logic: Apple and Banana is the same as Banana and Apple.

Formula: The number of ways to choose r objects from n distinct objects:

$$C(n, r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Note: Combinations result in fewer counts than permutations because duplicates (e.g., AB vs BA) are removed/divided out.

Combinations: Numerical Examples

Example: [Team Selection] A manager needs to pick 3 engineers from a pool of 10 to go on a trip.

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$$C(10, 3) = \frac{10 \times 9 \times 8}{3 \times 2 \times 1} = 120 \text{ teams.}$$

Introduction to Probability Theory

Quantifying Uncertainty

1. Random Experiment & Sample Space

Random Experiment: A process where the outcome cannot be predicted with certainty.

Sample Space (S): The set of **all** possible outcomes.

Single Coin Toss:

$$S = \{H, T\}$$

$$|S| = 2$$

Rolling a Die:

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$|S| = 6$$

The elements of the Sample Space follows the *MECE Principle*: Mutually Exclusive (outcomes don't overlap) and Collectively Exhaustive (cover all possibilities).

2. Defining Probability

Probability of an Event A: For a finite sample space where all outcomes are equally likely (Count-based):

$$P(A) = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}} = \frac{|A|}{|S|}$$

$$0 \leq P(A) \leq 1$$

Probability Examples (Count Based)

Example 1: [Tossing Two Coins] What is the probability of getting **at least one Head**?

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- $S = \{HH, HT, TH, TT\} \implies |S| = 4$.
- Favorable outcomes $A = \{HH, HT, TH\}$.
- $P(A) = 3/4 = 0.75$.

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Example 2: [Rolling Two Dice] What is the probability that the **sum of the dice is 7**?

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Example 2: [Rolling Two Dice] What is the probability that the **sum of the dice is 7**?

- Total outcomes: $6 \times 6 = 36$.
- Favorable pairs (x, y) : $\{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$.
- Count = 6.
- $P(\text{Sum} = 7) = 6/36 = 1/6 \approx 0.167$.

3. Kolmogorov's Axioms of Probability

The foundation of modern probability.

- ① **Non-negativity:** For any event E , $P(E) \geq 0$.
- ② **Normalization:** The probability of the sample space is 1. $P(S) = 1$.
- ③ **Additivity:** If A and B are mutually exclusive (cannot happen together), then:

$$P(A \cup B) = P(A) + P(B)$$

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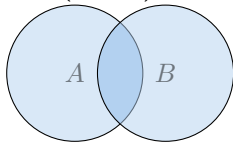
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Useful Corollaries:

- **Complement Rule:** $P(A^c) = 1 - P(A)$.
- **Subset Rule:** If $A \subset B$, then $P(A) \leq P(B)$.
- (If A is part of B , A cannot be more probable than B).

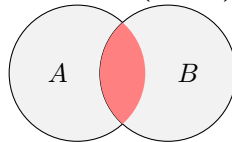
4. Events and Venn Diagrams

Union ($A \cup B$): OR



Any shaded area

Intersection ($A \cap B$): AND

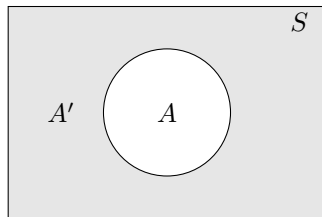


Only the overlap

Complementary Events

Complement (A' or A^c): NOT A

$$P(A') = 1 - P(A)$$



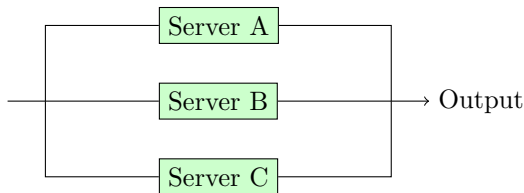
Physically: If probability of system failure is 0.01, reliability is 0.99.

5. Real World Example: Parallel Redundancy

Problem: Three servers (A, B, C) are connected in **Parallel**. System fails only if **ALL** servers fail.

- $P(\text{Fail}_A) = 0.1$
- $P(\text{Fail}_B) = 0.1$
- $P(\text{Fail}_C) = 0.1$

Find the probability of System Success.



Step 1: Find Probability of Total Failure Assuming failures are independent:

$$P(\text{System Fail}) = P(F_A) \times P(F_B) \times P(F_C)$$

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$$P(\text{System Fail}) = 0.1 \times 0.1 \times 0.1 = 0.001$$

Step 2: Find Probability of Success (Complement Rule)

$$P(\text{Success}) = 1 - P(\text{System Fail})$$

$$P(\text{Success}) = 1 - 0.001 = 0.999$$

Conclusion: Redundancy improves reliability from 90% (single server) to 99.9%.

Quick Check: Probability Fundamentals

Quiz: Probability

Q1: [Defective Items]. A box has 5 Good and 3 Bad parts. You pick 2 parts at random (without replacement). What is the probability **both are Bad**?

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- Ways to pick 2 Bad: $C(3, 2) = 3$.
- **Ans:** $3/28 \approx 0.107$.

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Q2: [At Least One]. A shooter hits the target with $P = 0.4$. If he shoots 3 times, what is the probability he hits the target **at least once**?

- Use Complement. $P(\text{Miss}) = 1 - 0.4 = 0.6$.
- $P(\text{Miss All}) = 0.6 \times 0.6 \times 0.6 = 0.216$.
- **Ans:** $1 - 0.216 = 0.784$.

Conditional Probability

Conditional Probability

Definition

Conditional probability is the likelihood of an event occurring, assuming that another event has **already occurred**.

Notation

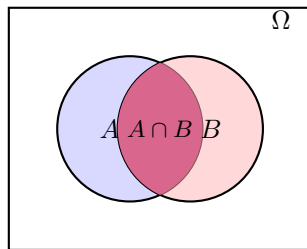
$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Read as: “Probability of A **given** B.”

Visualizing Conditional Probability

The Intuition:

- The original sample space is everything (the white box).
- Once we know B has happened, our “world” shrinks.
- The new sample space is just the circle B .
- We look for the portion of A that is **inside** B .



Why Learn This? ML Applications

Conditional probability is the engine behind many ML algorithms.

① Classification (Naive Bayes):

- Given an email contains the word “Free”, what is the probability it is Spam?
- $P(\text{Spam} \mid \text{“Free”})$

② Natural Language Processing (NLP):

- Next Word Prediction (LLMs like GPT).
- Given the previous word is “Artificial”, what is the prob the next word is “Intelligence”?
- $P(\text{“Intelligence”} \mid \text{“Artificial”})$

③ Recommendation Systems:

- Given a user bought a Laptop, what is the probability they buy a Mouse?
- $P(\text{Mouse} \mid \text{Laptop})$

Conditional Probability Ex 1: The “Tech Company” Scenario

Scenario: In a company of 100 employees:

- 40 are Data Scientists (DS).
- 25 employees use Python.
- 15 are Data Scientists AND use Python.

Question: If we pick a random employee and see they are a Data Scientist, what is the probability they use Python?

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Solution:

- 1 Event A : Uses Python. Event B : Is a Data Scientist.
- 2 $P(B) = 40/100 = 0.4$.
- 3 $P(A \cap B) = 15/100 = 0.15$.
- 4 Formula:

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{0.15}{0.40} = \mathbf{0.375}$$

Conditional Probability Ex 2: Rain and Traffic

Scenario:

- Probability of Rain (R) = 20%.
- Probability of Heavy Traffic (T) = 35%.
- Probability of Rain AND Heavy Traffic = 15%.

Question: You look out the window and see it is raining. What is the probability you will encounter heavy traffic?

Conditional Probability Ex 2: Rain and Traffic

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- Probability of Heavy Traffic (T) = 35%.
- Probability of Rain AND Heavy Traffic = 15%.

Question: You look out the window and see it is raining. What is the probability you will encounter heavy traffic?

Solution:

$$P(T \mid R) = \frac{P(T \cap R)}{P(R)}$$

$$P(T \mid R) = \frac{0.15}{0.20} = \mathbf{0.75}$$

Interpretation: Rain significantly increases the chance of traffic (from 35% to 75%).

Multiplication Rule

We can rewrite the conditional probability formula as:

$$P(A \cap B) = P(A) \cdot P(B \mid A)$$

Generalizing to n Events (The Chain Rule): For events $A_1, A_2, \dots, A_n$:

$$P(A_1 \cap \dots \cap A_n) = P(A_1) \cdot P(A_2 \mid A_1) \cdot \dots \cdot P(A_n \mid A_1 \dots A_{n-1})$$

This is fundamental in sequential decision making (like Reinforcement Learning).

Multiplication Rule Ex 1: Drawing Cards

Problem: You draw 3 cards from a deck of 52 without replacement. What is the probability they are all Aces?

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Solution: Let A_i be the event the i -th card is an Ace.

$$\begin{aligned} P(A_1 \cap A_2 \cap A_3) &= P(A_1) \cdot P(A_2 \mid A_1) \cdot P(A_3 \mid A_1 \cap A_2) \\ &= \frac{4}{52} \times \frac{3}{51} \times \frac{2}{50} \\ &= \frac{1}{13} \times \frac{1}{17} \times \frac{1}{25} \\ &\approx \mathbf{0.00018} \end{aligned}$$

Independent Events

Two events are **Independent** if knowing one occurred gives us **no information** about the other.

Definition

Events A and B are independent if:

$$P(A \mid B) = P(A)$$

Or equivalently:

$$P(A \cap B) = P(A) \cdot P(B)$$

Independence vs. Disjoint (Mutually Exclusive)

Confusion often arises here. Let's distinguish them using a Card Deck.

Disjoint (Mutually Exclusive)

- **Example:** Event A (Card is a King), Event B (Card is a Queen).
- Can they happen at the same time?
No.
- $P(A \cap B) = 0$.
- If I see a King, I know for a fact it is **not** a Queen. (Highly Dependent!)

Independent

- **Example:** Event A (Card is a King), Event B (Card is Red).
- Can they happen at the same time?
Yes (Red King).
- $P(A) = 4/52$.
 $P(A \mid \text{Red}) = 2/26 = 4/52$.
- Knowing the card is Red doesn't change the odds of it being a King.

Numerical Example: Disjoint vs Independent

Experiment: Rolling a fair 6-sided die.

Case 1: Disjoint Events

- $A = \{1, 2\}$, $B = \{5, 6\}$.
- $P(A) = 1/3$, $P(B) = 1/3$.
- $P(A \cap B) = 0$.
- Check Independence: Does $P(A \cap B) = P(A)P(B)$?
- $0 \neq 1/9$. **Not Independent.**

Case 2: Independent Events

- $A = \{1, 2\}$ (Low number), $C = \{2, 4, 6\}$ (Even number).
- $P(A) = 2/6 = 1/3$. $P(C) = 3/6 = 1/2$.
- Intersection: $A \cap C = \{2\}$. So $P(A \cap C) = 1/6$.
- Check Independence: Does $P(A \cap C) = P(A)P(C)$?
- $1/6 = (1/3) \times (1/2)$. **Yes, Independent.**

Quick Check: Conditional Probability

Quiz Q1: Independent or Not?

Problem: You roll a red die and a blue die.

- Event A: The sum of the dice is 7.
- Event B: The red die shows a 4.

Are A and B independent? Calculate to prove.

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Answer:

- $P(B) = 1/6$.
- Combinations for Sum 7: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1). Total $6/36 = 1/6$. So $P(A) = 1/6$.
- Intersection (Sum is 7 AND Red is 4): Only one case (Red 4, Blue 3). $P(A \cap B) = 1/36$.
- Check: $(1/6) \times (1/6) = 1/36$.
- **Yes, they are Independent!** Knowing the Red is 4 does not change the probability of the sum being 7.

Quiz Q2: Sequence Prediction

Problem: A user visits a website.

- If they are on the Home Page, 50% go to “Products”.
- If they are on “Products”, 30% go to “Buy”.

What is the probability a random user lands on Home, goes to Products, and then Buys?

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What is the probability a random user lands on Home, goes to Products, and then Buys?

Answer: This is a Chain Rule application.

$$P(H, P, B) = P(H) \times P(P \mid H) \times P(B \mid P)$$

Assuming everyone starts at Home ($P(H) = 1$):

$$1 \times 0.50 \times 0.30 = \mathbf{0.15} \text{ or } 15\%$$

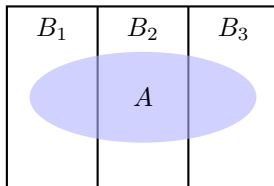
Total Probability and Bayes' Theorem

Theorem of Total Probability

If events $B_1, B_2, \dots, B_n$ partition the sample space (they are disjoint and cover everything), then for any event A :

$$P(A) = \sum_{i=1}^n P(A \mid B_i) \cdot P(B_i)$$

Why? We break a complex event A into smaller, manageable pieces based on the conditions B_i .



Total Probability Ex 1: Stock Market

Scenario: A stock's performance depends on the general economy state.

- Economy States: Bull (40%), Bear (30%), Stagnant (30%).
- Prob of Stock Rising in Bull market: 80%.
- Prob of Stock Rising in Bear market: 20%.
- Prob of Stock Rising in Stagnant market: 50%.

Q: What is the overall probability the stock rises?

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- Prob of Stock Rising in Stagnant market: 50%.

Q: What is the overall probability the stock rises?

Solution:

$$\begin{aligned}P(Rise) &= P(R \mid Bull)P(Bull) + P(R \mid Bear)P(Bear) + P(R \mid Stag)P(Stag) \\&= (0.8 \times 0.4) + (0.2 \times 0.3) + (0.5 \times 0.3) \\&= 0.32 + 0.06 + 0.15 \\&= \mathbf{0.53}\end{aligned}$$

Total Probability Ex 3: Factory Production

Scenario: A factory has 3 machines making chips.

- Machine 1: Makes 50% of chips, 1% defect rate.
- Machine 2: Makes 30% of chips, 2% defect rate.
- Machine 3: Makes 20% of chips, 3% defect rate.

Q: What is the total probability of picking a defective chip?

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Q: What is the total probability of picking a defective chip?

Solution: Let D = Defective.

$$\begin{aligned}P(D) &= P(D \mid M_1)P(M_1) + P(D \mid M_2)P(M_2) + P(D \mid M_3)P(M_3) \\&= (0.01)(0.50) + (0.02)(0.30) + (0.03)(0.20) \\&= 0.005 + 0.006 + 0.006 \\&= 0.017\end{aligned}$$

Total defect rate is **1.7%**.

Bayes' Theorem

Bayes' Theorem allows us to “flip” conditional probabilities. It updates our belief about an event based on new evidence.

$$P(A | B) = \frac{P(B | A) \cdot P(A)}{P(B)}$$

Expanded using Total Probability:

$$P(A | B) = \frac{P(B | A)P(A)}{P(B | A)P(A) + P(B | \neg A)P(\neg A)}$$

$\neg A$ is same as A^c , which means complementary event.

Terminologies

- $P(A)$: **Prior** (What we thought before evidence / data).
- $P(B | A)$: **Likelihood** (How probable is the evidence / data given our hypothesis?).
- $P(B)$: **Evidence** (Total probability of the evidence / data).
- $P(A | B)$: **Posterior** (Updated belief after seeing data / evidence).

Bayes' Theorem Ex 1: Employee Retention

Scenario:

- Probability an employee leaves the company this year ($P(L)$): 10%.
- If an employee is leaving, there is a 60% chance they logged in remotely late at night ($P(N | L) = 0.6$).
- If an employee is STAYING, there is only a 10% chance they logged in late at night ($P(N | \neg L) = 0.1$).

Q: We observe a user logging in late at night. What is the prob they are leaving?

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Q: We observe a user logging in late at night. What is the prob they are leaving?

Solution:

$$P(L|N) = \frac{0.6 \times 0.1}{(0.6 \times 0.1) + (0.1 \times 0.9)} = \frac{0.06}{0.06 + 0.09} = \frac{0.06}{0.15} = \mathbf{0.40}$$

The probability jumps from 10% to 40%.

Bayes' Theorem Ex 3: The Spam Filter

Context:

- 40% of all email is Spam (S).
- 60% is Ham (Not Spam) (H).
- The word “Offer” appears in 25% of Spam emails.
- The word “Offer” appears in only 5% of Ham emails.

Q: An email arrives containing the word “Offer”. What is the probability it is Spam?

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Q: An email arrives containing the word “Offer”. What is the probability it is Spam?

Solution:

$$\begin{aligned}P(S \mid \text{“Offer”}) &= \frac{P(\text{“Offer”} \mid S)P(S)}{P(\text{“Offer”} \mid S)P(S) + P(\text{“Offer”} \mid H)P(H)} \\&= \frac{0.25 \times 0.40}{(0.25 \times 0.40) + (0.05 \times 0.60)} \\&= \frac{0.10}{0.10 + 0.03} \\&= \frac{0.10}{0.13} \approx \mathbf{0.77}\end{aligned}$$

The presence of the word “Offer” raises spam probability from 40% to 77%.

Bayes' Theorem Ex 4: Diagnostic Test

Context: A disease affects 0.1% of the population ($P(D) = 0.001$).

- Test True Positive Rate: 99% ($P(+ | D) = 0.99$).
- Test False Positive Rate: 2% ($P(+ | \neg D) = 0.02$).

Q: Patient tests positive. Probability they have the disease?

Bayes' Theorem Ex 4: Diagnostic Test

Context: A disease affects 0.1% of the population ($P(D) = 0.001$).

- Test True Positive Rate: 99% ($P(+ | D) = 0.99$).
- Test False Positive Rate: 2% ($P(+ | \neg D) = 0.02$).

Q: Patient tests positive. Probability they have the disease?

Solution:

$$\begin{aligned}P(D) &= 0.001, & P(\neg D) &= 0.999 \\P(+ | D) &= 0.99, & P(+ | \neg D) &= 0.02 \\P(D | +) &= \frac{0.99 \times 0.001}{(0.99 \times 0.001) + (0.02 \times 0.999)} \\&= \frac{0.00099}{0.00099 + 0.01998} \\&= \frac{0.00099}{0.02097} \approx \mathbf{0.047} \quad (\approx 4.7\%) \end{aligned}$$

Result: *Even with a positive test, there is only a $\approx 5\%$ chance the person has the disease!*

Quick Check: Total Probability & Bayes

Quiz Q1: The “Fraud” Alert

Problem: A credit card fraud system detects 100% of real frauds, but also flags 1% of legitimate transactions as fraud (false alarm). Real fraud is rare: 0.1% of all transactions. If the system flags a transaction, what is the probability it is actually fraud?

Quiz Q1: The “Fraud” Alert

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Answer:

$$\frac{1.0 \times 0.001}{(1.0 \times 0.001) + (0.01 \times 0.999)} = \frac{0.001}{0.001 + 0.00999} \approx \frac{0.001}{0.011} \approx \mathbf{9\%}$$

Most flags are false alarms!

Quiz Q2: The Urn Challenge

Problem: Urn A has 2 Red, 1 Blue balls. Urn B has 1 Red, 2 Blue balls. You flip a fair coin. Heads $\rightarrow$ Pick from Urn A. Tails $\rightarrow$ Pick from Urn B. You picked a **Red** ball. What is the probability you flipped Heads (picked from Urn A)?

Quiz Q2: The Urn Challenge

Problem: Urn A has 2 Red, 1 Blue balls. Urn B has 1 Red, 2 Blue balls. You flip a fair coin. Heads $\rightarrow$ Pick from Urn A. Tails $\rightarrow$ Pick from Urn B.

You picked a **Red** ball. What is the probability you flipped Heads (picked from Urn A)?

Answer (Bayes):

- Prior: $P(Heads) = 0.5$.
- Likelihoods: $P(Red | Heads) = 2/3$. $P(Red | Tails) = 1/3$.

$$P(H|R) = \frac{(2/3 \times 1/2)}{(2/3 \times 1/2) + (1/3 \times 1/2)} = \frac{2/6}{2/6 + 1/6} = \frac{2}{3} \approx \mathbf{0.66}$$

Summary & Next Steps

Summary

- **Counting Principles:** The building blocks — multiplication, addition, permutations and combinations — for measuring “how many”.
- **Probability Fundamentals:** Kolmogorov’s axioms, events, and the complement rule.
- **Conditional Probability:** Updating probability based on new information.
- **Independence:** A crucial assumption for simplifying ML models.
- **Bayes’ Theorem:** The mathematical foundation for inference.

Next Steps

Next we shall learn about Probability Distributions (Discrete and Continuous).

Thank You!

Next Steps: Probability Distributions (Discrete and Continuous).